

Shareable link:

<https://studio.edgeimpulse.com/public/1006118/live>

Part 1:

```
Timing: DSP 79 ms, inference 552 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  Idle: 0.000000
  Lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 83 ms, inference 525 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  Idle: 0.996094
  Lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 84 ms, inference 523 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  Idle: 0.000000
  Lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 79 ms, inference 545 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  Idle: 0.996094
  Lift: 0.000000
Starting inferencing in 2 seconds...
|
```

Question 1:



Question 2:

The notebook compresses with pruning, strip wrappers, QAT, and int8.

The pruning is applied with `prune_low_magnitude` and uses a polynomial decay schedule with initial sparsity of .1, final sparsity of .8 and begin step of 0 and end step of 500.

After pruning, we used strip_pruning to remove the pruning wrappers.

QAT is applied after pruning using quantized_model().

The notebook uses additional mask-enforcement step during and after QAT instead of using a simple pruning-then-QAT pipeline to keep the pruned weights at zero so the model remains sparse.

The final model is converted into an int8 TFlite model using tf.lite.optimize.default, tflite_builtins_int8. The representative data is needed because we need the converter to correctly estimate the correct input and activation ranges for quantization.

Pruning and int8 quantization can help when deploying by reducing the number of active weights and reducing 32-bit floating-point values into 8bit integers.

Question 3:

The output is acceptable for resting board placed flat on a stable.

There some issues, some of the predictions were not precise, but according to the board, I can see why it appeared the way it labeled. Some causes for these issues/fluctuations were me slightly moving the board.

One way to change this deployment could be operating the Arduino in the same manner as the testing operations.

```
Final prediction: Lying down
Class index: 2
Confidence score: 0.4414
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Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.9961
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0430, 0.0000, 0.1055, 0.0664, 0.7656, 0.0000, 0.0000, 0.
Min-max-scaled model scores: 0.0000, 0.1172, 0.6875, 0.0000, 0.0547, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.1
Meta-classifier scores: 0.0078, 0.0117, 0.0078, 0.0039, 0.0039, 0.0078, 0.0117, 0.0078, 0.8984, 0.0117, 0.0078, 0.0195
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Final prediction: Cycling
Class index: 8
```

Question 4:

I did see some unusual prediction behavior after deployment. Sometimes it would incorrectly say laying down while I was actively moving the board. This can happen due to some differences in the testing sensor between my board and the testing board. Another factor that can affect the mislabeling could be different motion conditions during the testing that differ from the deployment.